Comparing ResNet and Plain CNN for Image Classification on CIFAR-10

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I. Introduction

Image classification is one of the most important tasks in computer vision because it is used in many applications such as object recognition, monitoring systems, and image-based automation. Convolutional Neural Networks (CNNs) are widely used for this task because they can learn useful image features directly from raw data. As CNNs became deeper, they often achieved better performance, but after a certain point, plain networks start getting worse because they become too hard to train and sometimes give worse results. To deal with this issue, residual networks (ResNet) introduced shortcut connections that let information move more easily through the network, making the learning process much smoother and made deeper models easier to optimize and helped improve image classification performance. In this project, we want to see how much these shortcuts help. We will build a ResNet-18 and a plain CNN-18 to compare how each model handles the CIFAR-10 dataset better using different tests like accuracy, precision, recall, F1-score, and confusion matrix.

II. Related Work

A. Early architectures

Early convolutional neural networks showed that deep networks could perform very well, but they were still quite simple in how they stacked layers and that they did not directly solve the problem of training much deeper networks [4]. Later, models like GoogLeNet tried to make networks better by using more complex designs called Inception to improve accuracy while keeping computation more efficient [5].

B. ResNet paper

The ResNet paper is the most important part of our project. The authors found that deeper plain networks can become harder to train when adding too many layers and may stop learning or even perform worse. They solved this by adding skip connections which are shortcut paths that help information to move more easily through the network allowing the network to keep the original information while learning new details [1].

C. CIFAR datasets

Some later studies compared ResNet-based models with other CNN architectures on image datasets such as CIFAR-10 and CIFAR-100. One study on CIFAR-10 compared an AlexNet-based model with ResNet-based model and the study eventually reported that a ResNet-based model achieved higher test accuracy [2]. Another study on CIFAR-100 compared ResNet-18, GoogLeNet, and VGG-16 and found that ResNet-18 is the best overall model [3]. These studies support the idea that residual connections can improve learning and classification performance.

D. Later architectures influenced by ResNet

Later architectures continued the general idea of improving information flow between layers such as DenseNet which connected each layer to all previous layers instead of using only additive skip connections showing that the idea introduced by ResNet had an important influence on later CNN architectures [6].

III. Novelty Statement

The original ResNet paper showed that residual connections help deep neural networks train better and improve image classification results [1]. In our project we are doing a focused comparison using the 18-layer version ResNet-18 and a plain CNN-18 on CIFAR-10. Instead of only looking at the final score (accuracy), we are using extra tools like F1-scores and Confusion Matrices. This will provide a clearer and more interpretable evaluation of whether residual connections improve classification performance.

IV. Methodology

A. How ResNet Works

In a plain CNN, each layer tries to learn everything from scratch, i.e. the direct mapping from the input x to the output $H(x)$, so $y = H(x)$. This means the stacked layers transform the input into a new representation. When the network becomes deeper, learning this direct mapping can become harder. In ResNet, the layers only have to learn the "residual" (the difference) between the input and the output. So the stacked layers learn the residual function instead of the full direct mapping. The residual function is written as: $F(x) = H(x) - x$ so the final output becomes: $y = F(x) + x$, where x is the shortcut that carries the original information forward. This shortcut prevents the network from forgetting what it already knows as it gets deeper.

B. Proposed Extension -Our Comparison Plan

For our extension, we build two models: a ResNet-18 and a plain CNN-18. Both models follow a similar 18-layers structure, but the main difference is that ResNet-18 has shortcuts (residual) connections, while the plain CNN-18 does not. The purpose is to make the comparison more straightforward and easier to interpret. We test and show this difference clearly on CIFAR-10. We are evaluating the models using multiple metrics to help showing which model performs better overall, and to also show how well each model handles class-level predictions and errors.

C. Dataset

We are using CIFAR-10. It has 60,000 color images divided into 10 different classes such as airplanes, cars, birds, cats, and others. We will split these into 50,000 for training and 10,000 for testing.

Before training the models, the CIFAR-10 images are converted into tensors. During training, simple data augmentation was also used through random horizontal flipping. These techniques help increase the diversity of the dataset and improve the generalization ability of the models.

D. Evaluation Metrics

The models will be evaluated using the following metrics:

- Accuracy: measures the total percentage of correctly classified images.
- Precision and Recall: measures if the model is biased toward certain classes.
- F1-score: gives a balanced measure between precision and recall.
- Confusion matrix: shows which images the model is getting mixed up.

V. Implementation Environment

The implementation was done in Google Colab using Python programming language. We mainly used PyTorch in addition to TorchVision, which is a library that works with PyTorch and is useful for our ResNet model building since it already provides a ready standard model so we didn't have to build every layer from scratch.

TorchVision was used in two main ways. First, it was used to load the CIFAR-10 dataset and apply the image preprocessing steps before training. Second, it was used to get the standard ResNet-18 model.

For the plain CNN model, we built our own CNN-18 model in PyTorch using standard layers such as convolution layers, batch normalization, ReLU, and a final fully connected layer. Fro the ResNet side, we used the TorchVision ResNet-18 implementation instead of manually writing the whole model from the referenced GitHub repository since that repository was more educational, while TorchVision gave us a clearer and easier starting point for the actual experiment.

Since CIFAR-10 images are small (32x32), we adjusted the default TorchVision ResNet-18 to make it more suitable for this dataset since the original ResNet-18 is built more for bigger images. In particular we changed the first convolution layer, removed the early max-pooling layer, and changed the final output layer so that the model predicts 10 classes instead of a larger number of classes. Both models were then trained from scratch (no pretrained weights) under the same general setup.

For evaluation and visualization, we used scikit-learn to calculate the metrics such as accuracy, precision, recall, F1-score, and confusion matrix. Also, we used Matplotlib to plot the training loss and test accuracy curves.

VI. Preliminary Results

As a first experiment, both models were trained for 5 epochs on CIFAR-10. The preliminary comparison results are shown in Table I. We noticed that even with this short training, ResNet-18 performed better than the plain CNN-18 across all reported metrics.

The training loss and test accuracy curves in Figures 1 and 2 show that both models were learning during the 5 epochs. However, ResNet-18 decreased faster in training loss and stayed higher in test accuracy than the plain CNN-18.

Since these are only preliminary results, they are not final yet. We still plan to continue the training for more epochs and report the final results later.

TABLE I
Preliminary Results After 5 Epochs on CIFAR-10

Model	Accuracy	Precision	Recall	F1-score
Plain CNN-18	68.85%	71.83%	68.85%	69.47%
ResNet-18	82.89%	83.98%	82.89%	83.01%

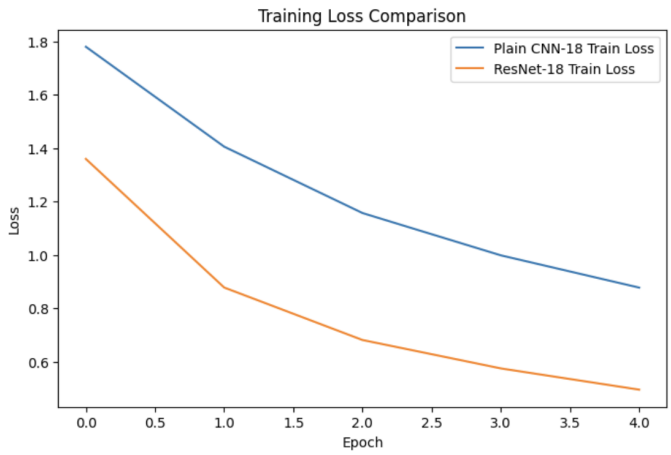

Fig. 1. Training loss comparison between Plain CNN-18 and ResNet-18 over 5 epochs.

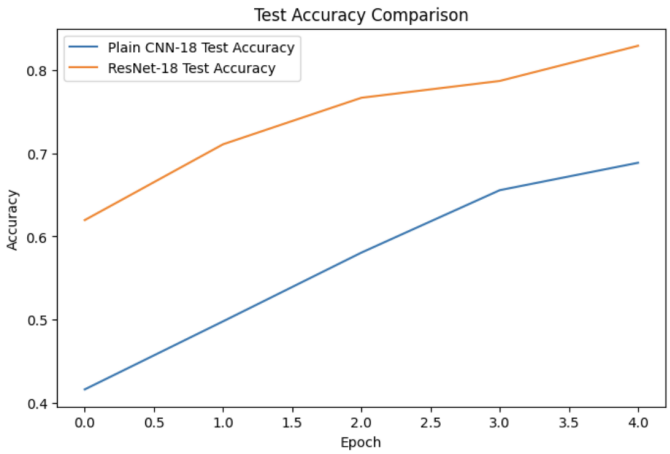

Fig. 2. Test accuracy comparison between Plain CNN-18 and ResNet-18 over 5 epochs.

VII. Issues Encountered

One issue we faced was how to make the comparison as fair as possible. At first, the default ResNet-18 settings were not the best fit for CIFAR-10, because the images are small. To solve this, we adjusted the first convolution layer, removed the early max-pooling layer, and changed the final layer to output the correct number of classes.

Another issue was deciding whether we should use the same implementation as the reference GitHub repository from phase 0. After checking, we found that the repository was more useful as a reference for understanding the paper and its implementation idea, while TorchVision was easier and clearer for actually running the experiment.

References

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